

1. Unit

Every physics quantity = magnitude $\times$ unit.

e.g. time lapsed = 15 minutes

student number = 100
 $\downarrow$
unit is one

► SI units - International System of Units (metric system)

• 7 base units

will be involved in this semester	quantity	SI unit
	length	m
	mass	kg
	time	s
	temperature	K
	electric current	A
	amount of substance	mol
	luminous intensity	cd

- All others can be derived from these 7 base units

► Dimension analysis

- useful for checking complicated equations
- convert all unit into the 7 base SI units
- only same unit can $+$ or $-$
e.g. $7\text{ m} - 4\text{ m} = 3\text{ m}$ is fine,
but $7\text{ m} + 3\text{ kg}$ doesn't make sense, thus not allowed.
- when $\times$ or $\div$, the "dimension" increases or decreases.
e.g. $3\text{ m} \times 3\text{ m} = 9\text{ m}^2$

- operate on the units, regardless of the numbers.
in the end, two sides of the equation should have same unit or called "dimension".

Very useful!

► Conversion of units

e.g. convert 5 m/s into miles per hour?

$$(1 \text{ mile} = 1609 \text{ m}, \quad 1 \text{ hr} = 3600 \text{ s})$$

★ hint: treat units as part of the equation.
You can do $\times$ and $\div$ on units

I inserted a 1 here.

$$\begin{aligned}
 5 \frac{\text{m}}{\text{s}} &= 5 \frac{\text{m}}{\text{s}} \left(\frac{\text{mile}}{\text{hr}} \cdot \frac{\text{hr}}{\text{mile}} \right) \\
 &= 5 \cdot \frac{\text{m}}{\text{mile}} \cdot \frac{\text{hr}}{\text{s}} \cdot \frac{\text{mile}}{\text{hr}} \\
 &= 5 \cdot \frac{1}{1609} \cdot 3600 \cdot \frac{\text{mile}}{\text{hr}}
 \end{aligned}$$

organize

$$\begin{aligned}
 1 \text{ mile} = 1609 \text{ m} &\Rightarrow \frac{\text{m}}{\text{mile}} = \frac{1}{1609} \\
 1 \text{ hr} = 3600 \text{ s} &\Rightarrow \frac{\text{hr}}{\text{s}} = 3600
 \end{aligned}$$

$$\approx 11.2 \text{ mile/hr}$$

► prefix of unit

prefix	symbol	multiply factor
...	...	...
giga	G	10^9
mega	M	10^6
kilo	k	10^3
		1
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
...	...	...

e.g. 1 kg = 1000 g

1 dm = 0.1 m

* Review of scientific notation

e.g. $3456 = 3.456 \times 10^3$

↳ tells you the order of magnitude

→ tells you the exact amount
(only leave one digit in front of decimal point)
(here there are four significant digits)

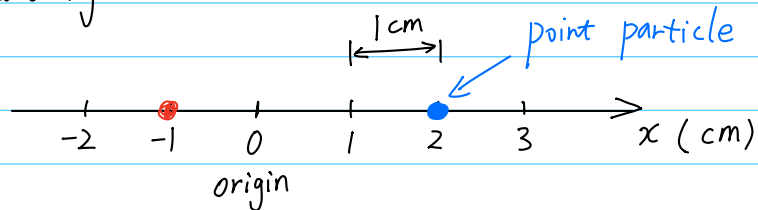
e.g. $2.5 \times 10^{-5} \text{ m} = 2.5 \times 10 \text{ } \mu\text{m} = 2.5 \times 10^4 \text{ nm}$

These are all appropriate scientific notations.

2. Motion in One Dimension

2.1. Position, average velocity, average speed

► coordinate system



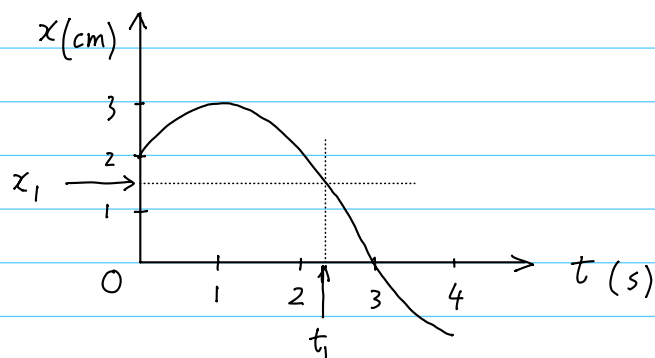
position $x = 2 \text{ cm}$ 2 cm to the positive direction

$x = -1 \text{ cm}$ 1 cm to the negative direction, or
-1 cm to the positive direction.

Describe the motion of the particle by expressing the position as a function of time.

$$x = f(t)$$

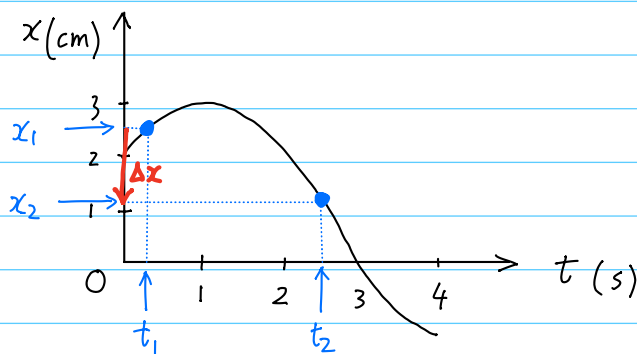
e.g.



- To find out the position of the particle at a specific time t_1 , take a vertical cut at $t = t_1$, the cut intersects with the curve project the intersection to the position axis, that tells you
At $t = t_1$, $x = x_1$
- As time evolves, the particle moves $\updownarrow$ in this graph, but it's actually moving $\leftrightarrow$ in the x -axis.

► displacement and distance

Let's consider two time points t_1 and t_2 .



displacement $\Delta x \equiv x_2 - x_1$ final position minus the initial position
 "is defined as"

- if $x_2 > x_1$, $\Delta x > 0$, the particle moved to the right
- if $x_2 < x_1$, $\Delta x < 0$, the particle moved to the left
- displacement has both magnitude $|x_2 - x_1|$ and a direction left/right
 $\therefore$ displacement is a vector -/+

distance $d \equiv$ the total length of the path
 d always ≥ 0
 distance is a scalar

e.g., Find the displacement and distance from 0s to 3s.

At $t_1 = 0s$, $x_1 = 2cm$. At $t_2 = 3s$, $x_2 = 0cm$.

Displacement: $\Delta x = x_2 - x_1 = -2cm$
 (2cm to the left)

Distance: $\left. \begin{array}{l} \text{from } 0s \text{ to } 1s, \text{ distance } 1cm \\ \text{from } 1s \text{ to } 3s, \text{ distance } 3cm \end{array} \right\} \text{ total } 4cm.$

► average velocity and speed

average velocity

$$v_{\text{avg}} \equiv \frac{\Delta x}{\Delta t}$$

- same direction as Δx
- is a vector
- SI unit m/s

average speed

$$v_{\text{avg}} \equiv \frac{d}{\Delta t}$$

- always ≥ 0
- is a scalar
- SI unit m/s

For the e.g. above, average velocity $v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{-2 \text{ cm}}{3 \text{ s}} \approx -0.67 \text{ cm/s}$

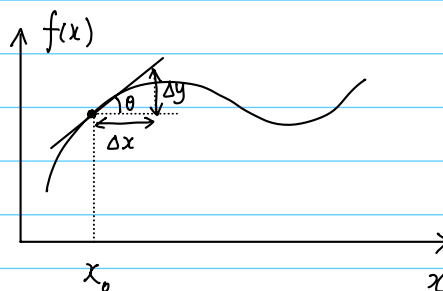
average speed $v_{\text{avg}} = \frac{d}{\Delta t} = \frac{4 \text{ cm}}{3 \text{ s}} \approx 1.33 \text{ cm/s}$

2.2 Instantaneous velocity and speed

► review of derivative in math

Given a function $y = f(x)$,

the derivative of f with respect to x at $x = x_0$ is the slope of the tangent line at $x = x_0$.



$$f'(x) = \tan \theta$$

To get this $\tan \theta$, we consider an infinitesimally small triangle at this point

$$\tan \theta = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

$$f'(x) = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

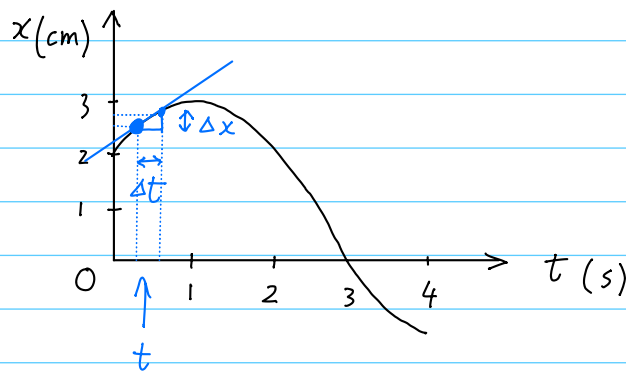
► instantaneous velocity

Let the variable be t , and the function be the position vs time.

$$\frac{f(t + \Delta t) - f(t)}{\Delta t} \Rightarrow \text{average displacement in duration } \Delta t$$

Let $\Delta t \rightarrow 0$

$$\lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \Rightarrow \text{the instantaneous velocity at moment } t.$$



$$v \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

- v is a vector, has direction (+ - sign of slope)
- We might omit the word "instantaneous" from now on!
- The sign of velocity is independent of the sign of position.

► From velocity to displacement

$$\Delta x = \int_{t=t_1}^{t=t_2} v(t) dt = x(t_2) - x(t_1)$$

$$\text{or } x(t_2) = \int_{t=t_1}^{t=t_2} v(t) dt + x(t_1)$$

► instantaneous speed $\equiv |v|$, is a scalar